Assignment 1:

Question:

For comparing the cellular structures and chemical compositions of two different animal tissues using gram staining, hematoxylin staining, wet mount, dry mount, paper chromatography, thin layer chromatography (TLC), and spectrophotometry:

1.Line up the given techniques in order of application during whole procedure.

For comparing cellular structures and chemical compositions of two different animal tissues, the techniques you listed would typically be used in the following order:

1. Wet mount - This technique is often the initial step for viewing the basic cellular structures under a microscope. It involves placing the tissues in a liquid medium on a slide.
2. Dry mount - This is similar to a wet mount but involves no liquid medium. It may be used alongside or as an alternative to wet mount depending on the tissue type and the focus of observation.
3. Gram staining - This staining technique allows for the differentiation of bacteria present in the tissues, if that is of interest. It's not typically used for animal tissues, but can be applied if microbial analysis is needed.
4. Hematoxylin staining - This is a common histological stain used on animal tissues to visualize cellular components, particularly nuclei, under a microscope.
5. Spectrophotometry - This method can be used to assess the concentration of certain chemicals in the tissues. It would be used after the tissues have been homogenized and the relevant molecules extracted.
6. Paper chromatography - This technique can be applied to separate and identify different compounds from the tissue extracts. It's simpler but less precise than TLC.
7. Thin layer chromatography (TLC) - Following paper chromatography, TLC might be used for a more detailed analysis of chemical compounds in the tissue extracts.

This sequence assumes a comprehensive analysis starting from basic microscopy to more complex chemical identification methods.

2. List out the materials required for whole procedure.

For the comprehensive analysis of cellular structures and chemical compositions of animal tissues using the methods you listed, you would need the following materials:
1. Microscope – For viewing slides prepared during wet and dry mount, as well as those stained with gram and hematoxylin.
2. Slides and cover slips – For preparing wet and dry mounts.
3. Staining racks and staining jars – For holding the slides during the staining processes.
4. Gram staining kit – Typically includes crystal violet, iodine solution, alcohol decolorizer, and safranin.
5. Hematoxylin stain – For staining cell nuclei.

6. Eosin stain (optional for counterstaining in hematoxylin staining) – Provides contrast by staining other cell components.
7. Distilled water – Used in various steps for rinsing and preparations.
8. Tissue homogenizer or blender – For breaking down the tissues for chemical analysis.
9. Centrifuge – For separating particulates from solutions, if necessary.
10. Spectrophotometer – For measuring the absorbance of light by the extracted chemicals.
11. Cuvettes – For holding samples in the spectrophotometer.
12. Chemicals for extraction – Such as buffers, solvents, and acids or bases as needed based on the target molecules.
13. Paper chromatography supplies:
- Chromatography paper
- Solvent system (depending on the compounds of interest)
- Developing chamber

14. Thin layer chromatography (TLC) supplies:
- TLC plates (usually coated with silica gel or alumina)
- TLC developing chamber
- Appropriate solvents for the mobile phase

15. Pipettes and pipettors – For transferring precise volumes of liquids.
16. Gloves and safety goggles – For personal protection during handling of chemicals and biological samples.
17. Analytical balance – For measuring small quantities of chemicals.
18. pH meter or strips – To ensure proper pH of solutions, if needed.

This extensive list covers the broad requirements for the various techniques involved, from microscopic examination to chemical analysis.

3. What will be the conclusion of whole procedure?

The conclusion of the procedure involving various techniques to compare the cellular structures and chemical compositions of two different animal tissues would generally include the following insights:

1. Structural Differences:

- **Microscopy Observations**: Wet and dry mounts, along with hematoxylin and gram staining, will reveal the basic cellular structures, differences in cell size, shape, and arrangement between the two tissues. Hematoxylin staining will highlight the nuclei and possibly other cell structures, providing a clear visualization of the cellular architecture.
- Staining Results: Differences in how tissues respond to gram staining (if applicable) might indicate varying microbial presence or structural properties. Hematoxylin and eosin staining will further differentiate cellular components, making it possible to compare tissue complexity and organization.

2. Chemical Composition:

- **Spectrophotometry**: This will provide quantitative data on specific compounds or groups of compounds within the tissues, such as proteins, nucleic acids, and lipids, based on their absorbance of light at different wavelengths.
- **Chromatography (Paper and TLC):** These techniques will separate and possibly identify different biochemical molecules present in the tissue extracts. The patterns of separation can indicate the presence of unique or differing compounds between the tissues.

- **Comparative Analysis**: Differences in the chromatographic profiles and spectrophotometry readings can be used to infer functional variations between the tissues, such as metabolic activities or the presence of specific biomolecules.

3. Integrated Findings:

- Combining the structural and chemical data, the conclusion would likely discuss how the observed differences in cellular architecture and biochemical composition relate to the physiological roles and health status of the tissues. For instance, variations might correlate with different metabolic rates, susceptibility to diseases, or functional capacities of the tissues.

4. Applications and Implications:

- The findings might have implications for biomedical research, particularly in understanding disease mechanisms, developmental biology, or in the design of therapeutic interventions.
- The comparative approach can help in identifying biomarkers of diseases or in the understanding of normal versus pathological tissue states.

In sum, the conclusion of this procedure would provide a comprehensive view of how two different animal tissues compare on a microscopic and molecular level, potentially leading to broader biological and medical insights.